Do Now · retrieval

Lesson 078 · Binary Search: Bounds and Sorted Data

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Binary search precondition?

Q2

What does low represent?

Q3

What is mid for bounds low0 high6?

Q4

Binary search target15 in [1,3,5,7,9,11,13,15]: give midpoint indexes.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Use binary search only when elements are sorted; keep inclusive low and high candidate bounds.

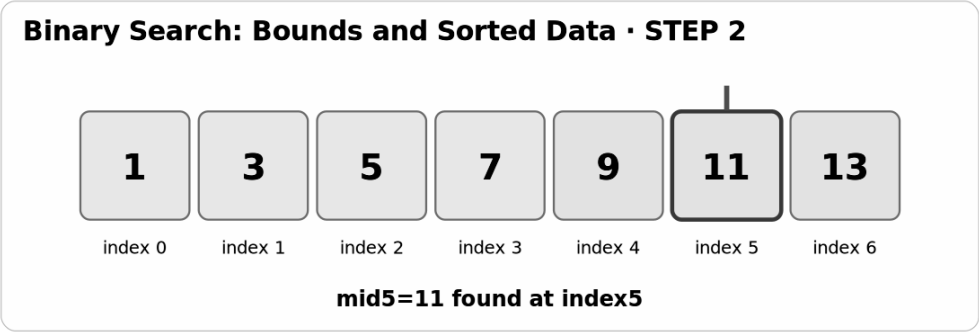
Core idea 2

Trace — At every mismatch discard the midpoint and half the candidates with low=mid+1 or high=mid−1.

Core idea 3

Explain — Each unsuccessful step shrinks the search interval. Worst time O(log n), iterative extra memory O(1).

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

low — Lowest candidate index.

high — Highest candidate index.

midpoint — Middle candidate.

halving — Repeatedly reducing interval by about half.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

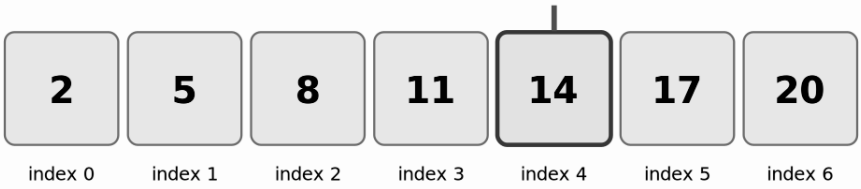
Find14 in [2,5,8,11,14,17,20].

Follow the worked method

1. Check the working — mid3=11, mid5=17, mid4=14, returns index4.

Source model · final illustrated step

IDO · TRACE



mid3=11, mid5=17, mid4=14, returns index4.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Check the working
mid2=5, mid0=1, mid1=3, low2>high1: return−1.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Trace target11 in [1,3,5,7,9,11,13]. [6]

Question 2

Write three binary-search comparison cases and correct bounds. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare iterative binary and linear search with respect to time and precondition. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

After a too-small midpoint, setting low=mid always ensures progress.

Use key vocabulary from memory

1. Explain low in your own words.

2. Explain high in your own words.

3. Explain midpoint in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.